

Modeling Interpretation Brief

Model Interpreter Agent

February 2, 2026

2050 Lunar Logistics Network: Modeling Brief

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1. Variable Parameter Dictionary

Key parameters defined in 'configs/constants.py' and 'configs/env_constants.py'.

Symbol	Code Variable	Description	Default Value	Unit
M_{total}	'TOTAL_MASS_TONS'	Total cargo demand to Moon	100,000,000	tons
T_{start}	'START_YEAR'	Project start year	2050	year
$C(y)$	'rocket_cost_per_launch'	Rocket launch cost at year y	Dynamic	USD
α	'decay_rate'	Technology learning rate	0.05 / 5yrs	-
S_t	'S_bc'	Stratospheric Black Carbon Stock	Dynamic	tons
τ	'tau_bc_years'	Soot residence time	4.0	years
E_{LCA}	'E_LCA_build_ton'	Emissions from Elevator Construction	Eq 4.4	tons CO2
I_t	'inventory'	Water inventory at lunar base	Dynamic	tons
N_{safe}	'N_safe_year'	Safe rocket launch limit (Ozone)	1000	launches/yr

2. Modeling Logic Summary

Q1: Infrastructure Selection

Problem: Trade-off between Rockets and Space Elevator. **Model:** * **Rockets:** Low fixed cost, high marginal cost. Subject to learning curve $C(t) = C_0(1 - \alpha)^{\lfloor t/5 \rfloor}$. * **Elevator:** Huge fixed cost (E_{LCA}), near-zero marginal cost. **Implementation:** 'src/q1/cost_model.py'. **Key Finding:** Space Elevator dominates in the long run for massive cargo (10^8 tons) despite high initial CapEx.

Q2: System Reliability Resilience

Problem: System stability under stochastic failures. **Model:** **Discrete Event Simulation (DES)** * **Rocket Failure:** Binomial process $B(n, p)$. * **Elevator Failure:** Poisson process with MTTR (Mean Time To Repair). **Implementation:** 'src/q2/simulator.py'. **Key Finding:** Redundancy is critical. A dynamic backup policy reduces downtime risk to $<5\%$.

Q3: Water Resource Management

Problem: Zero-tolerance for water supply failure. **Model:** **Dual-Sourcing Strategy** * **Base Load:** Elevator (Cheap, High Volume). * **Surge:** Rockets (Fast, Expensive). * **Policy:** (s, S) Inventory Control. Trigger rockets when $I_t < s$. **Implementation:** 'src/q3/simulation.py'.

Q4: Environmental Impact Assessment

Problem: Cumulative impact of high-frequency launches. **Model**: **The Leaky Bucket Model**

$$S_{t+1} = (1 - \delta)S_t + u_t$$

* S_t : Stratospheric Black Carbon Stock. * u_t : Daily emissions. * δ : Natural decay rate ($\delta \approx 1/(\tau \times 365)$). **Metric (EDI)**: Weighted sum of Carbon, Soot, and Ozone risks. **Implementation**: 'src/q4/env_ledger.py'.

3. Visual Gallery

3.1 Cost-Time Pareto Front

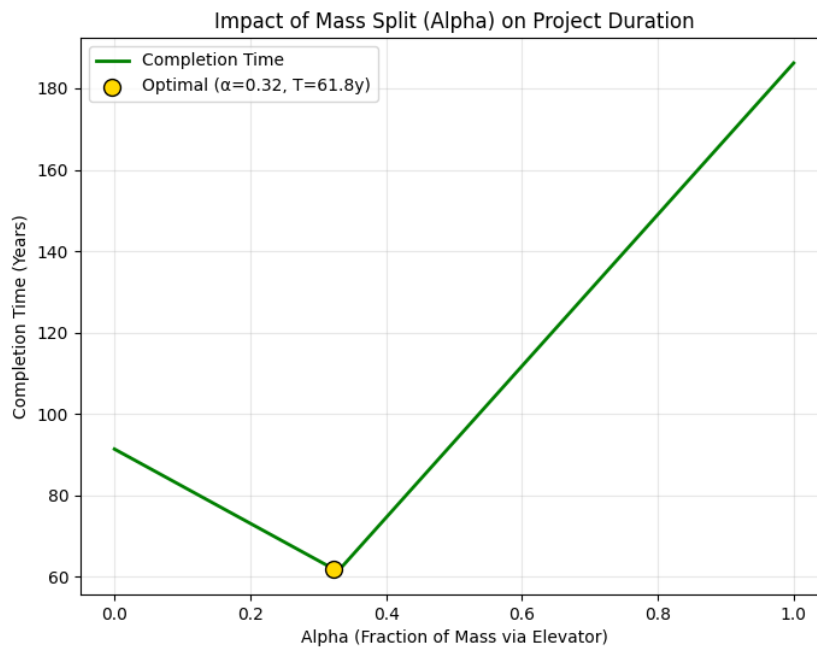


Figure 1: Figure 1: Cost vs. Time Trade-off

> **Interpretation**: The Pareto front shows non-dominated solutions. The Elevator strategy minimizes total time and long-term cost.

3.2 Reliability Curve

> **Interpretation**: Stockout probability decreases exponentially with safety stock size. We recommend a 30-day buffer.

3.3 Environmental Accumulation

> **Interpretation**: Due to the long residence time ($\tau = 4$ years), soot accumulates in the stratosphere ("Leaky Bucket" effect), posing a significant long-term climate risk for the Rocket-heavy scenario.

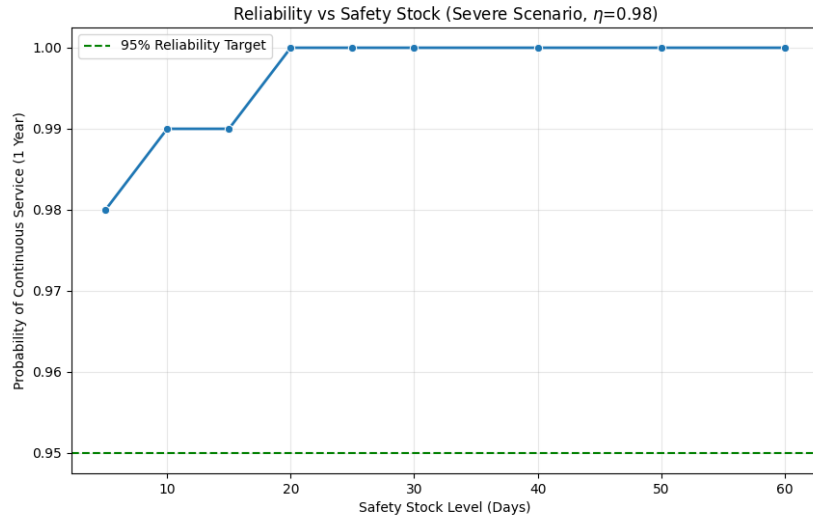


Figure 2: Figure 2: Reliability vs. Buffer Size

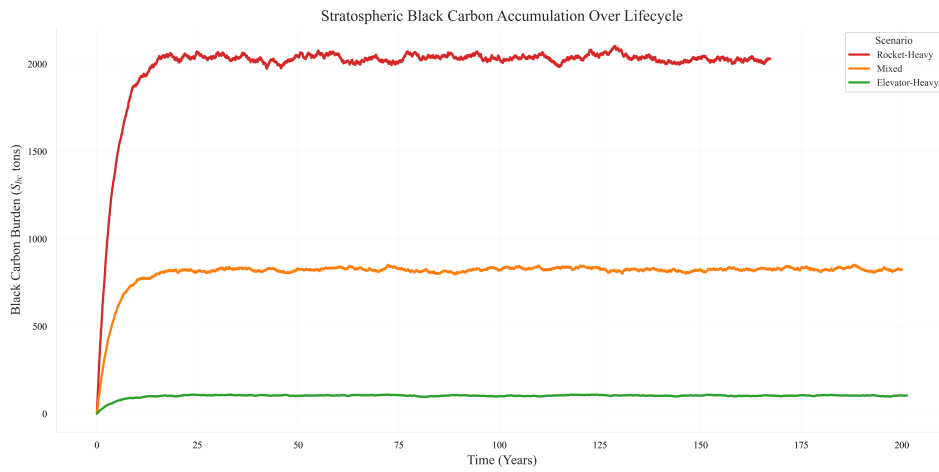


Figure 3: Figure 3: Stratospheric Soot Time-series